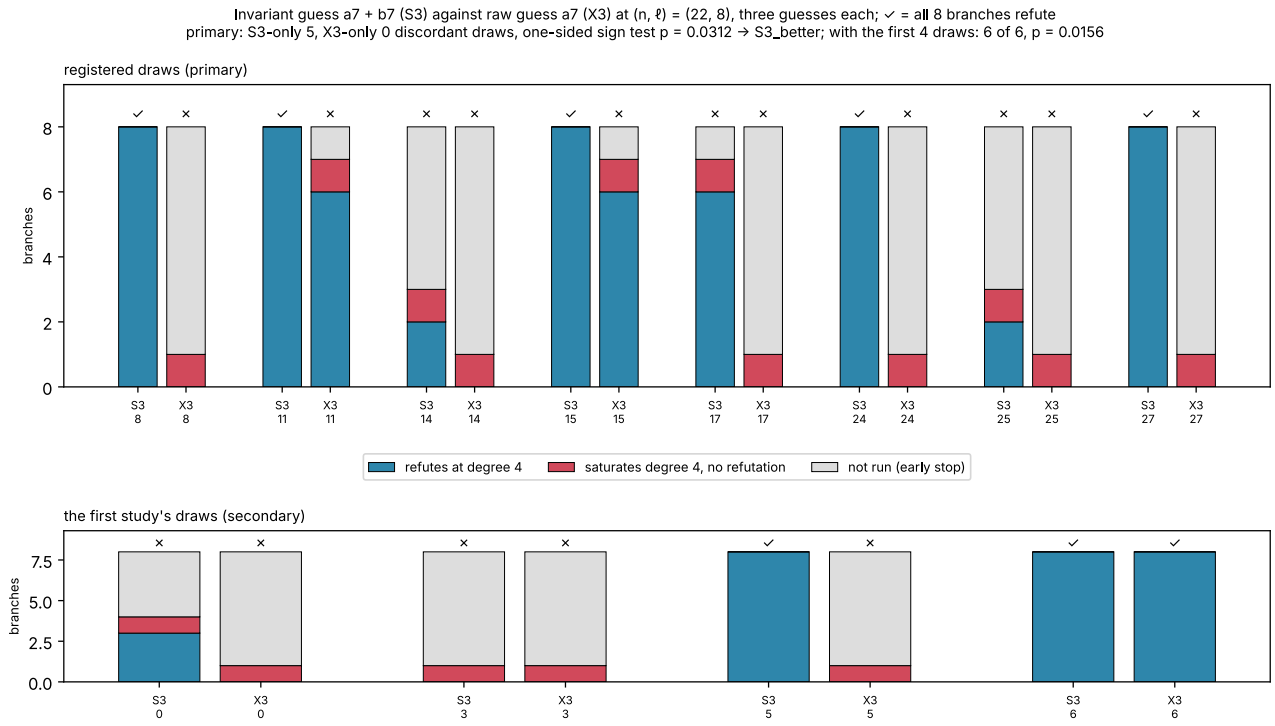
Guessing the swap invariant beats guessing a raw coordinate at $\ell = 8$ ($p = 0.031$)

Study `research/dreg_invariant_guess_more_20261029/`, pre-registered on fresh draws. **Class:** stage diagnostic. **Registered verdict:** S3_better.

Question

The guess-then-degree-4 oracle fixes variables until the MutantXL closure of the $m = 3$ chained S_3 system refutes at degree 4. At three guesses on the (22, 8) cell, does fixing the swap-invariant coordinate $a_7 + b_7$ of $s = x_1 + x_2$ (S3) succeed on more draws than fixing the raw a_7 (X3)?

Result



- On the 8 new draws, S3 suffices (every branch refutes) on 5 and X3 on 0.
- All 5 discordant draws favour S3: one-sided exact sign test $p = 0.031$.
- With the 4 draws of 2851: 6 of 6, $p = 0.016$. S3 suffices on 7 of 12 draws and X3 on 1 of 12.
- Controls valid: the replay check, the identity and exactness checks on every run, and the positive control on 8 of 8.

Reading

- The coordinate of the invariant `s`, not the dimension of the summands, governs refutation at this cell. S3 leaves both summands at dimension 8.
- In the guess-then-degree-4 oracle, invariant guesses are the measured better choice. This is a stage-level improvement over the index-calculus baseline.
- It moves no row and none beats rho: the guess count at the priced $\ell = 33$ cell is still extrapolated far above the break-even of 6.
- **Next:** two invariant guesses on the pair, $\ell = 9$, and $m = 4$.